

Q1. Create a dataframe using weather.csv containing values of Temperature, Humidity, Rainfall, and Wind Speed. Generate a Heat Map to visualize the correlation between these weather parameters.

[10 Marks]

Q2. Apply the K-Means Clustering algorithm to the given student dataset using Attendance and Marks as the input attributes. Perform the required data preprocessing, set $K = 3$, assign the students to the respective clusters, and visualize the clusters and their centroids using a suitable scatter plot.

[20 Marks]

OR

Q2. Create a suitable dataset named Loan_application.csv containing at least 10 records with the attributes Application_ID, Age, Income, Credit_Score, Loan_Amount, Employment_Years, Debt_Ratio, and Loan_Status, where Loan_Status represents whether the loan application is Approved or Rejected.

Write a Python program to perform the following data reduction techniques:

- a) Feature Selection: Identify the important features for predicting Loan_Status.
- b) Feature Extraction: Reduce the number of numerical attributes using a suitable technique such as Principal Component Analysis (PCA).

[20 Marks]

Q3. Viva

[5 Marks]